

Elbow Method for Clustering

Core Formula

The Elbow Method evaluates clustering quality using **Within-Cluster Sum of Squares (WCSS)** (also called **Sum of Squared Errors (SSE)** or **Inertia**). The formula is:

$$WCSS/SSE = \sum_{i=1}^k \sum_{x_j \in C_i} |x_j - \mu_i|^2$$

k: Number of clusters

C_i: Points in cluster *i*

x_j: A data point in *C_i*

μ_i: Centroid of *C_i*

x_j − *μ_i*²: Squared Euclidean distance between point *x_j* and centroid *μ_i*

Key Insight: As *k* increases, SSE decreases. The "elbow point" (optimal *k*) occurs where SSE's rate of decrease sharply slows.

Numeric Example

Dataset: 8 points in 2D space:

[(1, 1), (1.5, 1.8), (5, 8), (8, 8), (10, 0.6), (9, 11), (0, 1), (3, 4)]

Step 1: Compute SSE for different *k*

<i>k</i>	Clusters (Example)	SSE Calculation	SSE
1	All points in one cluster	Centroid = (4.75, 5.55)SSE = Σ points - centroid ² = 285.23	285.23
2	Cluster 1: [(1,1), (1.5,1.8), (0,1), (3,4)]Cluster 2: Other points	Centroid ₁ = (1.375, 1.95)SSE ₁ = 12.5Centroid ₂ = (8, 6.9)SSE ₂ = 28.7Total SSE = 12.5 + 28.7	41.2
3	K-means output (e.g., 3 groups)	SSE ≈ 18.3	18.3
4	K-means output (e.g., 4 groups)	SSE ≈ 8.6	8.6

Step 2: Plot SSE vs. k

k	1	2	3	4
SSE	285.23	41.2	18.3	8.6

Step 3: Identify the Elbow Point

Sharp drop: From $k = 1$ to $k = 2$ (SSE ↓ by 244.03).

Slower decrease: From $k = 2$ to $k = 3$ (↓ 22.9), $k = 3$ to $k = 4$ (↓ 9.7).

Elbow: At $k = 2$ (sharpest change in slope).

Conclusion: Optimal number of clusters = 2.

Advanced: Precise Elbow Detection

If visual inspection is unclear, compute the angle ψ_k at each k :

$$\tan\psi_k = \frac{-SSE(k+1) + 2 \cdot SSE(k) - SSE(k-1)}{1 + (SSE(k) - SSE(k-1)) \cdot (SSE(k+1) - SSE(k))}$$

The k with **smallest** $\tan\psi_k$ (sharpest angle) is the elbow point.

Key Notes

1. **Terminology:** WCSS, SSE, and Inertia are synonymous.
2. **Limitations:**
 - Elbow point may not align with true clusters (e.g., penguin species data where $k = 2$ is chosen over true $k = 3$).
 - Subjective if SSE decreases smoothly.
3. **Best Practice:** Use with domain knowledge and other metrics (e.g., silhouette score).